

PROJECT SYLORA 2

Forensic Audit Report

Baseline state assessment · 2026-08-13

Evidence-based forensic audit

SYLORA — Full Forensic Audit (Source of Truth)

Project: Project Sylora 2 (SYLORA)

Audit date: 2026-08-13

Auditor: Cloud forensic pass (code + runtime + screenshots)

Branch baseline: `main` @ forensic run

Rule: Claims require evidence. README and prior agent reports were not trusted without verification.

Related documents

Document	Path
Architecture (as-is)	SYLORA ARCHITECTURE MAP.md
UI map	SYLORA UI MAP.md
Real vs Mock	SYLORA REAL VS MOCK.md
Duplication	SYLORA DUPLICATION REPORT.md
Security	SYLORA SECURITY AUDIT.md
Production readiness	SYLORA PRODUCTION READINESS.md
Remediation roadmap	SYLORA REMEDIATION PLAN.md
Screenshots	/workspace/audit/screenshots/desktop/ , mobile/

1. Executive summary

SYLORA is a **monolithic Node.js application** serving a **vanilla JS SPA** (~897 LOC frontend, ~6500 LOC backend core+ecosystem) with **296 HTTP endpoints**, optional **PostgreSQL/Redis**, and **OpenAI** integration. The product presents a **broad surface area** (social, live, AI, business OS, science, gifts) but **most capabilities are UI + API stubs** backed by JSON file storage in dev.

Completion (weighted, honest)

Metric	%
Overall SYLORA completion	27%
UI completion	68%
Functional completion	22%
Backend completion	48%
End-to-end completion	14%
Production readiness	18%

Module counts

Category	Count
Verified 100% working (strict E2E)	4
Partial	45
Mock/static	12
Broken	1
Missing	15
P0 blockers	4
P1 issues	12

2. What was run (evidence log)

Check	Result	Evidence
npm install	PASS	59 packages, 0 vulns
npm run lint	PASS	exit 0
npm run build	PASS	syntax-check only
npm test	134/134 PASS	DATABASE_URL= REDIS_URL= empty
Server node src/server.mjs	PASS	:8787
GET /api/health	ok, json-dev-runtime	runtime curl
Register → post → gift → live	PASS	node runtime script
AI chat	503 AI_PROVIDER_NOT_CONFIGURED	expected without key
Docker compose	NOT RUN	docker not installed
Browser screenshots	22 captured	audit/screenshots/
WebRTC live/calls	NOT VERIFIED	requires browser media
Postgres production mode	NOT VERIFIED	no DB in audit env

3. Inventory (repository map)

See [SYLORA_ARCHITECTURE_MAP.md](#).

Applications: 1 (Node server + static SPA)

Packages: root only (`sylora@0.1.0`)

SDK: `sdk/js/` minimal

Database: `infra/postgres/schema.sql` + migrations 002-012

Docker: Dockerfile + compose.yaml

CI/CD: none

Tests: 134 files in `tests/`

Assets: ~45MB PNG in `public/assets/`

Deprecated: old `docs/audit/`, `renderProfileLegacy`, patch scripts

4. UI / navigation audit

Full per-page detail: [SYLORA UI MAP.md](#).

Navigation issues found

Issue	Severity
/videos , /gifts , /ai not in left rail (by design but hidden)	P3
Duplicate wallet surfaces (header, profile, gifts)	P2
LIVE "Following" always empty	P2
Admin only via More grid	OK
Locale selector offers 14 langs; server PATCH only uk/pl/en	P2
No Google sign-in button	P1 product gap

Proposed unified IA (not implemented)

Home | Live | Create▼ | Discover | Messages | Sylora | Profile
Settings▼ → Identity, Security, Business tools, Developer, Admin

5. Button interaction matrix

Control	Status	Evidence
Register (API)	WORKING	POST 201 runtime
Register (UI form)	PARTIAL	Same API; not browser-automated
Login	WORKING	API 401 on bad creds
Logout	WORKING	API 200
Google auth	MISSING	No UI/route
Profile edit	PARTIAL	PATCH /api/me exists
Follow/like/comment	PARTIAL	API exists
Language selector	WORKING	onchange → setLocale + PATCH (app.js account())
Messages/calls	PARTIAL	API; WebRTC BLOCKED
Start/join stream	PARTIAL	Create API OK; media BLOCKED
Gifts send	WORKING	POST 201 TEST LUMEN
Wallet/payments	MOCK	TEST label
Subscriptions	MISSING	—
AI interaction	BLOCKED	No OPENAI_API_KEY
Settings	WORKING	Navigation

6. Responsive audit

Viewport	Score	Notes
Mobile 360–412	68/100	Dock OK; tall heroes
Tablet	NOT VERIFIED	—
Desktop 1366–1920	72/100	Sidebar layout works

Issues: overflow rare; touch targets generally OK; modals not fully tested; keyboard safe areas partially via CSS `viewport-fit=cover`.

7. Design system audit

Verdict: Light futuristic SYLORA aesthetic **is present** (aurora, glass cards, horizon PNG) but **fragmented across 11 CSS files** with overlapping tokens.

Element	Consistent?
Typography	Mostly (Inter-like stack in styles.css)
Colors/gradients	Yes — warm light theme
Spacing/cards	Similar radius ~28px heroes
Dark dashboard drift	No — stays light
Icons	Mix emoji + unicode symbols
Loading/skeleton	Minimal spinners
a11y	Partial ARIA; contrast generally OK on light bg

Pages that feel like different products: **none severe** — consolidation CSS unified scenes per `data-view`.

8. Sylora AI audit

Capability	Status
Provider	OpenAI via <code>openai</code> npm ^7.4
Models	<code>OPENAI_MODEL</code> default gpt-5.6; realtime gpt-realtime-2.1
Backend	server.mjs + ecosystem/service.mjs
System prompts	personalityFor + contextPack per view
Tool calling	3 tools, 3-round max
Memory	JSON/PG; user + ai_confirmed sources
Streaming text	Via Responses API output_text (not SSE to client)
Voice	OpenAI Realtime HTTP SDP exchange
STT/TTS	Browser Speech + OpenAI transcription config
Multilingual	UI i18n; AI replies in user language (prompt)
Business/creator/education assistants	Ecosystem personas — PARTIAL
Live co-host	living-sylora + hooks — BLOCKED without AI
Moderation	Safety filter patterns in living-sylora
Rate limit	12/min/user AI
Fallback	503 when no key; lexical search fallback
Cost controls	trackAiUsage in ecosystem — not verified

Readiness	%
AI UI	75%
AI backend	55% (code) / 0% live without key
Voice	40% code / BLOCKED
Memory	70%
Live co-host	15%

9. Living Sylora / Avatar audit

Feature	Reality
Model format	PNG sprite sheets (not GLB/GLTF)
Renderer	DOM + CSS + <code></code> layers
Skeleton/rig/blendshapes	NOT SUPPORTED BY CURRENT MODEL
Blinking/gaze	CSS classes + JS timers — REAL
Lipsync	CSS viseme grid from audio bands — FAKE/SIMULATION
Emotions	PNG atlases exist; limited use in AI hero
Voice sync	Partial viseme frames
Realtime 3D	MISSING
Mobile performance	Heavy PNG loads (~2MB base avatar)

Classification: **CSS/PNG/SPRITE — NOT real 3D avatar.**

10. Live / streaming audit

Feature	Class
UI	UI exists
Create room	Functional locally (API verified)
WebRTC mesh	Prototype (code complete, not E2E tested)
RTMP/OBS ingest	Missing (companion is control bridge only)
Chat/likes/gifts	Functional locally (API + SSE)
Battles	Partial
Recording	Prototype (canvas capture in studio)
TURN	BLOCKED_EXTERNAL
Production-ready	No

11. Live gifts audit

Catalog: 10 gifts in `/api/gifts` (spark...infinite-sylora). Gift V2 passports: 20 named in `gift-v2/catalog.js`.

Gift tier	Renderer	Sound	Backend tx
basic-legendary v1 IDs	GPU Three.js procedural OR 2D atlas	gift-sfx.js	REAL in JSON; atomic in PG tests
phoenix-rebirth v2	WebGL + keyframe PNGs	physical-audio	Same send pipeline

Not a video gift system — interactive WebGL/Canvas with optional LIVE segmentation (**BLOCKED** without MediaPipe bundle).

12. Authentication audit

Flow	Result
Registration	PASS API
Login	PASS
Logout	PASS (even without token)
Session refresh	MISSING — single long-lived bearer
Recovery/verify	MISSING
Google/phone	MISSING
Protected routes	Client-side renderAuth + server 401
Admin	env email list → role admin
scrypt passwords	PASS (auth.mjs)
Token storage	localStorage — XSS risk

13. Backend API table

296 endpoints documented in architecture map. Summary:

Area	Endpoints	Frontend uses
Core social/live	~90	app.js
Ecosystem	~206	~140 via app.js, 5 command-palette, 1 obs-overlay
Backend-only	~150	No UI caller

Full METHOD|PATH table: `grep src/server.mjs + src/ecosystem/routes.mjs` or see subagent export in git history.

Status: Most return JSON; **live verification** only on subset without Postgres/OpenAI.

14. Database audit

Area	JSON dev	Postgres schema
Users/sessions	yes	users, sessions
Social	yes	posts, reactions, comments
Messages	yes	conversations, messages
Live	yes	live_rooms, live_messages + migration 012 state
Wallet/gifts	yes	wallets, ledger, gifts
AI	yes	ai_messages, ai_memories, ai_actions
Ecosystem	JSON blobs in store	migrations 010-011 ecosystem

Schema drift risk: server still writes many entities only to JSON store even when auth social PG enabled (communities, courses, videos, reports).

15. Performance

Item	Finding
public/assets	45MB
app.js	897 LOC single file — no minification
11 CSS files	Sequential blocking
Three.js vendor	~1MB+
Lazy loading	Minimal
Gift GPU	Can stress mobile GPU — quality governor exists
ffmpeg	Sync spawn blocks event loop during transcode

16. Accessibility

Check	Status
Keyboard nav	Partial — buttons focusable
Focus visible	CSS partial
Labels	Some aria-label
Contrast	Generally OK light theme
Touch targets	Mobile dock OK
Reduced motion	Respected in scheduleSyloraLife
Forms	Basic labels missing on some prompts

17. Test coverage truth

134 tests PASS — but they **do not** mean product-ready:

Layer	Covered?
Unit (parsers, gifts, motion)	Yes
API integration (in-memory server)	Yes — strong vertical slices
Postgres (pg-mem)	Yes
E2E browser	No
WebRTC	No
AI live OpenAI	Mock server only
Security penetration	No
Load	No

Critical flows **without** E2E tests: live viewer, calls, payments, OAuth, AI voice.

18. User journeys

Journey	Result	Break point
NEW: landing → register → home	PARTIAL	Home renders; onboarding MISSING
RETURNING: login → nav → profile	PARTIAL	API OK; full UI not browser-tested
AI: open → message → history	FAIL	503 no OpenAI key
CREATOR: studio → go live	BLOCKED	Camera/WebRTC not audited
VIEWER: discover → join → gift	PARTIAL	Gift send API OK; watch BLOCKED
SOCIAL: follow → message → call	PARTIAL	Calls BLOCKED
MONETIZATION: wallet → purchase	FAIL	TEST LUMEN only; no payments

19. Architectural problems (6–12 month risk)

1. **God files:** app.js, ecosystem/service.mjs — untestable UI/regression risk
2. **Dual persistence:** JSON + Postgres branches everywhere
3. **296 endpoints / 21 views:** massive backend-without-frontend debt
4. **In-process ffmpeg:** won't scale
5. **SSE fanout in memory:** won't scale without Redis (designed but optional)
6. **TEST LUMEN** misrepresents monetization
7. **Avatar marketing vs PNG reality** — trust risk
8. **No CI** — regressions undetected

20. TODO / placeholder scan

Signal	Count	Notes
renderProfileLegacy	1	dead
comingSoon i18n key	UI string only	
PAYMENT_PROVIDER	env blocked	
OAuth OAUTH_DOC	architectural	
mock in tests	legitimate	
Following tab empty	honest placeholder	

21. Weighted readiness by domain

Domain	%	Rationale
Frontend shell	72	All views render
Design/UI	65	Beautiful but fragmented CSS
UX	45	Many dead-end modules
Responsive	60	Mobile OK; tablet unverified
Backend	48	Wide API; many untested live
Database	40	Schema exists; dual-write
Authentication	35	Email/pass only
Sylora AI	12	Blocked without key
Avatar	20	PNG simulation
Voice	15	Blocked
Live streaming	25	API yes; WebRTC unproven
Social	30	Core post OK
Messaging	25	API exists
Calls	20	Code only
Gifts	40	Send OK; playback heavy
Wallet/payments	8	Mock currency
Creator tools	35	Studio UI + API
Business	30	JSON/ecosystem
Education	30	Free courses OK
Notifications	25	SSE exists
Analytics	10	Minimal
Admin	25	JSON reports
Security	42	Baseline headers/auth
Performance	35	Heavy assets
Testing	38	Many unit, no E2E
DevOps	15	No CI
Production infra	18	Compose only

22. VERIFIED 100% WORKING (strict)

Only features with UI + backend + runtime verification:

1. **Static SPA shell + routing** (21 views resolve, server fallback)
2. **Health / ready probes**
3. **Email/password register + login** (JSON mode, scrypt, bearer token)
4. **Create text post + list feed** (API runtime)

Everything else missing at least one of: browser E2E, Postgres mode, external provider, or media permissions.

23. Candidates for deletion (do not delete in audit)

See [SYLORA_DUPLICATION_REPORT.md](#).

Top candidates: `renderProfileLegacy`, unused expression PNG v1, old audit markdown claims, redundant design CSS after merge, backend-only experimental endpoints without product owner.

24. Missing as a product (strategic)

- Real identity (OAuth, verify email)
- Working AI as differentiator (currently off)
- Proven live streaming with discovery loop
- Real money loop (not TEST LUMEN)
- Trust & safety ops (reports in JSON, no workflow)
- Recommendation/discovery (search only)
- Network effects (Following live empty)
- Mobile apps
- Compliance (delete account, DPA)
- Observability and on-call

25. P0 / P1 classification

P0 — Blockers

1. No CI/CD
2. Production persistence not enforced
3. AI provider not configured (if AI is core promise)
4. WebRTC TURN missing for live product claim

P1 — Critical

- TEST LUMEN presented as currency
- Media GET without auth
- No Google auth / recovery
- Avatar capability misrepresentation risk
- 150 orphan APIs increase attack surface

26. Audit methodology notes

- Prior `audit/AUDIT_REPORT.md` claims about `owmWeather` / `getThree` **were false positives** — symbols not in repository.
- Language selector **does work** (contrary to interim browser report).
- Three.js: `importmap` + relative imports; vendor addons use bare `three` — works in modern browsers with `importmap`.

27. Change log (audit-only)

No product code modified. Added:

- docs/audit/SYLORA_*.md (this bundle)
- audit/screenshots/** (22 webp)
- audit/INDEX.md , SUMMARY.txt , AUDIT_REPORT.md (working notes)

End of source-of-truth audit. Next step: approve [SYLORA REMEDIATION PLAN.md](#) before any fixes.

Appendix: SYLORA_ARCHITECTURE_MAP

Source of truth derived from repository inspection + runtime checks. Not from README or prior agent reports.

Repository topology

Project-Sylora-2/	
— src/	# Node.js backend (ESM)
— server.mjs	# HTTP server, core API (~590 LOC)
— store.mjs	# JSON file persistence (dev fallback)
— auth.mjs	# scrypt password + bearer tokens
— companion.mjs	# OBS WebSocket bridge (separate process)
— infra/	# postgres.mjs, redis.mjs
— repositories/	# Postgres repos (auth, wallet, ai, live, ecosystem, outbox, cor
— ecosystem/	# Mega-module: service.mjs (~3885 LOC), routes.mjs (~1215 LOC),
— public/	# Static SPA + assets (~45MB PNG/SVG)
— index.html	# Shell + 11 CSS + importmap for Three.js
— app.js	# Entire frontend SPA (~897 LOC)
— gift-engine.js, gift-gpu-engine.js, gift-v2/*, gift-runtime.js	
— obs-overlay.html/js, phoenix-preview.html/js	
— assets/	# PNG atlases, avatar sprites, phoenix keyframes
— infra/postgres/	# schema.sql + 11 migrations (002–012)
— tests/	# 134 Node test files (*.test.mjs), no Playwright/Cypress
— scripts/	# migrate.mjs, deploy-prod.sh, patch scripts
— sdk/js/	# Minimal JS SDK stub
— compose.yaml + Dockerfile	
— docs/	# Design docs + prior audits (not authoritative)

Runtime architecture (actual)

```
flowchart TB
  Browser[Browser SPA public/app.js]
  Node[Node HTTP src/server.mjs]
  JSON[(data/sylora.json)]
  PG[(PostgreSQL)]
  Redis[(Redis)]
  OpenAI[OpenAI API]
  Companion[companion.mjs OBS bridge]

  Browser -->|REST + SSE| Node
  Node -->|if DATABASE_URL| PG
  Node -->|if no DATABASE_URL| JSON
  Node -->|if REDIS_URL| Redis
  Node -->|if OPENAI_API_KEY| OpenAI
  Browser -->|local WS optional| Companion
  Node -->|static files| Browser
```

Persistence modes

Mode	Trigger	Evidence
JSON dev runtime	DATABASE_URL empty	GET /api/health → persistence: json-dev-runtime (verified runtime)
Postgres hybrid	DATABASE_URL set	authSocial.enabled , wallet/ai/live repos; migrations in infra/postgres/migrations/
Ecosystem cache	Always	Large JSON arrays in store.data + optional PostgresEcosystemRepository

Production `compose.yaml` expects Postgres 17 + Redis 8 + migrate on boot.

API surface

Layer	Count	Files
Core routes	~90	src/server.mjs
Ecosystem routes	~206	src/ecosystem/routes.mjs
Total HTTP endpoints	296	Grep-verified
SSE streams	6	/api/events , /api/gifts/stream , /api/live/:id/events , /api/conferences/:id/events , /api/calls/:id/events , /api/studio/browser-source/events
WebSocket on /api/*	0	Signaling via HTTP POST + SSE fanout

Frontend architecture

- **Single-page app:** one `app.js` , pathname routing via `SPA_SHELL_VIEWS` (21 views).
- **No React/Vue/Svelte** — vanilla JS string templates.
- **No build step for UI** — `npm run build` only syntax-checks JS.
- **Design:** 11 stacked CSS files (v2–v6, consolidation, avatar, etc.).

Realtime / media

Feature	Implementation	Production-ready?
Live WebRTC	HTTP signaling + SSE; <code>LivePeerRegistry</code> + optional Redis	Functional locally; needs TURN for NAT
Calls / conferences	Same pattern via <code>call-engine</code> , <code>conference-fanout</code>	Code exists; browser E2E not verified in audit
Live chat / gifts	SSE fanout	Verified API + SSE endpoint exists
Video upload	Raw POST to <code>/api/media/upload</code> ; ffmpeg HLS in-process	Requires ffmpeg (present on audit VM)
OBS	<code>companion.mjs</code> + browser overlay token	Companion not started in audit
RTMP / CDN streaming	Not present	—

AI architecture

Component	Location	Status
Text chat	<code>runSyloraAi()</code> in <code>server.mjs</code> + OpenAI Responses API	BLOCKED without <code>OPENAI_API_KEY</code> (503 verified)
Tool calling	<code>get_my_context</code> , <code>propose_post</code> , <code>propose_memory</code>	Implemented; needs OpenAI
Voice realtime	POST <code>/api/ai/realtime</code> → OpenAI Realtime HTTP	BLOCKED without key
Memory	JSON or <code>postgres-ai</code> ; manual + confirmed actions	Works without OpenAI (CRUD verified)
Living Sylora / director	<code>ecosystem/living-sylora/</code> , platform events	In-memory + optional AI hook
Ecosystem intelligence	<code>ecosystem/service.mjs</code> (3000+ LOC)	Many endpoints; ~150 without frontend callers

Security boundaries (implemented)

- Bearer session tokens (SHA-256 hashed at rest in Postgres mode)
- Rate limits: in-memory + optional Redis (`allowRequest` , `allowAi`)
- CSP, X-Frame-Options, nosniff (`securityHeaders()`)
- Admin via `SYLORA_ADMIN_EMAILS` env
- Password: `script` (`auth.mjs`)

What is NOT in the architecture

- Separate frontend/backend repos
- Microservices (monolith)
- Mobile native apps
- CI/CD pipelines (no `.github/workflows`)
- Kubernetes / Terraform
- Real payment provider
- Google OAuth flow (env placeholders only)
- Email verification / password recovery
- CDN / object storage (media on local disk)

Deprecated / experimental signals

- `renderProfileLegacy()` in `app.js` — dead code, superseded by `renderProfile()`
- Multiple `scripts/patch-*.mjs` — one-off migration patches
- Prior `docs/audit/*.md` reports — superseded by this forensic baseline
- `docs/FINAL_IMPLEMENTATION_REPORT.md` — do not trust without verification

Test architecture

- **134 Node native tests** (`npm test` with empty `DATABASE_URL/REDIS_URL`)
- Covers: API E2E slices, postgres repos (pg-mem), gifts, motion, security headers, RTC config
- **No browser E2E** in CI sense; `tests/avatar-ui.test.mjs` checks file existence only

Appendix: SYLORA_UI_MAP

Evidence: `public/index.html` , `public/app.js` , screenshots in `audit/screenshots/` .

Navigation shell (persistent)

Element	Location	Routes / actions
Brand	Header	→ <code>/ feed</code>
Command palette ⌘K	Header <code>#globalSearch</code>	Search, slash commands → <code>explore/ai/messages/etc.</code>
Locale <code><select></code>	Header <code>#localeSwitch</code>	14 options; WORKING — <code>setLocale()</code> + <code>PATCH /api/me (app.js account())</code>
LUMEN balance ◇	Header (authed)	→ <code>/gifts</code>
Inbox ○	Header (authed)	→ <code>/messages</code>
Avatar	Header (authed)	→ <code>/profile</code>
Logout ⇄	Header (authed)	POST <code>/api/auth/logout</code>
Sign in	Header (guest)	→ <code>renderAuth()</code> inline
Left rail primary	<code>index.html</code>	feed, live, clips, studio, learning, business, explore, communities
Left rail secondary		messages, profile, more, create hub
Sylora mini	Left rail	→ <code>/ai</code>
Right rail		Live pulse, Sylora CTA
Mobile dock	Bottom	feed, live, ai, messages, profile

Screenshots: `audit/screenshots/desktop/01_home_feed.webp` , `audit/screenshots/mobile/*.webp`

SPA views (21)

1. Home — / → feed

Field	Value
Render	<code>renderFeed()</code>
Purpose	Personalized home, composer, carousels
APIs	<code>/api/feed</code> , <code>/api/live</code> , <code>/api/users</code> , <code>/api/communities</code> , <code>/api/courses</code> , <code>/api/businesses</code> , <code>/api/home/hub</code> , <code>/api/daily-brief</code> , <code>POST /api/posts</code>
Auth	Browse public; composer requires login
Status	PARTIAL — feed/composer work (API verified); hub/brief need auth
Mock	LUMEN labeled TEST
Screenshot	desktop <code>01_home_feed.webp</code> , mobile <code>01_home_mobile.webp</code>

2. LIVE — /live

| Tabs | discover, following, create, battles, studio link | | APIs | `/api/live`, `/api/live/entertainment`, WebRTC `/api/live/rtc-config`, SSE `/api/live/:id/events` | | Status | **PARTIAL** — room list/create verified API; WebRTC watch **BLOCKED** (no browser media test); following tab empty by design | | Screenshot | `02_live.webp`, `02_live_mobile.webp` |

3. Studio — /studio

| Purpose | Camera, canvas, scenes, OBS/companion, go live | | APIs | `/api/studio/scenes`, `POST /api/live`, browser-source token | | Auth | Required | | Status | **PARTIAL** — scenes API verified; camera/WebRTC **BLOCKED** in headless audit | | Screenshot | `04_studio.webp` |

4. Clips — /clips

| APIs | GET `/api/videos?format=clip`, upload `/api/media/upload`, `POST /api/videos` | | Status | **PARTIAL** — list works; upload needs file + ffmpeg | | Screenshot | `03_clips.webp` |

5. Videos — /videos

| Nav | Feed carousel, More → Media (not in left rail) | | Status | Same pipeline as clips — **PARTIAL** |

6. Explore — /explore

| APIs | `/api/search`, `/api/search/universal` (authed) | | Status | **PARTIAL** — lexical search works; semantic may degrade without AI | | Screenshot | `07_explore.webp` |

7. Messages — /messages

| Sub-tabs | messages, notifications, invites, calls, priority | | APIs | conversations, notifications, calls, SSE /api/events | | Auth | Required | | Status | **PARTIAL** — API layer exists; WebRTC calls **BLOCKED** in audit | | Screenshot | 09_messages.webp , mobile 04_messages_mobile.webp |

8. Sylora AI — /ai

| APIs | /api/ai/history , chat, realtime, memory, actions | | Auth | Required | | Avatar UI | PNG sprite + CSS visemes (sylora-avatar-v2-base.png , gestures) — **NOT 3D** | | Status | **PARTIAL** — UI renders; chat **BLOCKED** (no OPENAI_API_KEY); voice **BLOCKED** | | Screenshot | 10_ai.webp , mobile 03_ai_mobile.webp |

9. Profile — /profile

| APIs | /api/me , stats, progress, ledger, notifications; PATCH profile | | Status | **PARTIAL** — profile CRUD verified via API | | Screenshot | 11_profile.webp , mobile 05_profile_mobile.webp |

10. Gifts — /gifts

| APIs | GET /api/gifts , POST /api/gifts/send | | Status | **PARTIAL** — send works in TEST LUMEN (API verified); WebGL gift playback depends on GPU | | Screenshot | 12_gifts.webp |

11. More (Settings hub) — /more

| Purpose | Grid launcher to identity, agents, developer, security, dashboard, canvas, admin, videos, gifts | | Status | **REAL** navigation shell | | Screenshot | 13_more.webp , mobile 06_more_mobile.webp |

12. Identity — /identity

| APIs | /api/identity , /api/kg | | Auth | Required | | Screenshot | 14_identity.webp |

13. Agents — /agents

| APIs | /api/agents , install, negotiations | | Screenshot | 15_agents.webp |

14. Developer — /developer

| APIs | /api/developer/apps , keys; OAuth doc is static JSON | | Screenshot | 16_developer.webp |

15. Security — /security

| APIs | security-center, memory, privacy, reputation | | Not screenshotted | Auth required; same shell as other gated views |

16. Dashboard — /dashboard

| APIs | /api/dashboard , /api/ai/command | | Not screenshotted | |

17. Canvas — /canvas

| APIs | /api/canvas | | Not screenshotted | |

18. Communities — /communities

| Sub-views | openCommunity() , channel posts (no URL change) | | APIs | /api/communities , channels, social extensions | | Screenshot | 08_communities.webp |

19. Learning — /learning

| Sub-views | courses, science hub, conference WebRTC | | APIs | courses, /api/learning/* , /api/science/* , conferences | | Screenshot | 05_learning.webp |

20. Business — /business

| APIs | businesses, orgs, CRM, invoices, conferences | | Screenshot | 06_business.webp |

21. Admin — /admin

| Access | role === 'admin' only | | APIs | /api/admin/reports , audit | | Not screenshotted | Requires admin account |

Standalone pages (not SPA views)

Page	URL	Purpose	Status
Phoenix preview	/phoenix-preview.html	Gift V2 cinematic demo	REAL client-only WebGL
OBS overlay	/obs-overlay.html?token=	Stream overlay SSE	PARTIAL — needs valid token

Auth gate (inline, no route)

| Render | `renderAuth()` | | Forms | Register (username, email, password) / Login (identity, password) | | APIs | POST `/api/auth/register`, `/api/auth/login` | | Google | **MISSING** — no UI button | | Recovery | **MISSING** | | Status | **REAL** — register/login verified API + runtime |

Visual tree (actual product IA)

SYLORA

- └─ Home (/)
- └─ LIVE (/live)
 - └─ Discover
 - └─ Following [empty state]
 - └─ Create stream
 - └─ Battles
 - └─ → Studio (/studio)
- └─ Clips (/clips)
- └─ Studio (/studio)
- └─ Science & Learning (/learning)
 - └─ Courses [inline]
 - └─ Science conference [inline WebRTC]
- └─ Business (/business)
 - └─ Company directory
 - └─ Org workspace [inline]
 - └─ Business conference [inline]
- └─ Discover (/explore)
- └─ Communities (/communities)
 - └─ Channel view [inline]
- └─ Inbox (/messages)
 - └─ Messages / Notifications / Invites / Calls / Priority
 - └─ Call session [inline WebRTC]
- └─ Sylora AI (/ai)
- └─ Profile (/profile)
- └─ Gifts (/gifts) [header shortcut]
- └─ Videos (/videos) [hidden nav]
- └─ Settings hub (/more)
 - └─ Identity (/identity)
 - └─ Agents (/agents)
 - └─ Developer (/developer)
 - └─ Security (/security)
 - └─ Dashboard (/dashboard)
 - └─ Canvas (/canvas)
 - └─ Admin (/admin) [role gate]
- └─ Create Hub [modal overlay]
- └─ Command palette [modal overlay]

Responsive notes (screenshot audit)

Viewport	Score	Notes
Desktop 1440×900	72/100	Sidebar + content OK; 11 CSS files load
Mobile 390×844	68/100	Bottom dock works; hero sections tall
Tablet	Not captured	Same CSS breakpoints in <code>modules.css</code> — NOT VERIFIED

Console / network (verified vs false positives)

Issue	Verified?
favicon.ico 404	YES — no favicon in public/
AI degraded banner	YES — without OPENAI_API_KEY
<code>owmWeather</code> / <code>getThree</code> ReferenceError	NOT IN CODEBASE — false positive in interim agent report; not reproduced via static analysis
Three.js bare <code>from 'three'</code> in vendor addons	Mitigated by importmap in index.html; gift-gpu uses relative paths

Appendix: SYLORA_REAL_VS MOCK

Classification per module. Evidence: code + runtime/API tests on 2026-08-13.

Legend: **REAL** = end-to-end verified | **PARTIAL** | **MOCK** | **STATIC UI** | **PLACEHOLDER** | **BROKEN**
| **MISSING** | **BLOCKED**

Platform core

Module	Class	Evidence
HTTP server + static SPA	REAL	Server running :8787; all routes 200
JSON persistence mode	REAL	health → json-dev-runtime; register/post/gift API OK
Postgres persistence	PARTIAL	11 migrations + repos; not runtime-tested (Docker unavailable)
Redis fanout	PARTIAL	Code + tests with mocks; not runtime-tested
Rate limiting	PARTIAL	In-memory works; Redis path untested live
Session auth (email/password)	REAL	Register/login/logout API verified
Google OAuth	MISSING	<code>integrations.mjs</code> → BLOCKED_EXTERNAL; no UI
Phone auth	MISSING	No routes
Email verification	MISSING	No routes
Password recovery	MISSING	No routes
Admin moderation	PARTIAL	API exists; needs admin role to E2E

Social

Module	Class	Evidence
Feed + posts	REAL	POST/GET verified
Reactions, comments	PARTIAL	API exists; UI not browser-tested
Follow / block	PARTIAL	API exists
Notifications	PARTIAL	API + SSE; not browser-tested
Universal search (lexical)	REAL	<code>/api/search</code> works
Universal search (AI)	BLOCKED	Needs OpenAI / authed semantic endpoint
Communities	PARTIAL	API tested in <code>api.test.mjs</code>
Public profiles	STATIC UI	Letter avatars only

Messaging & calls

Module	Class	Evidence
DM conversations	PARTIAL	API in server.mjs; not browser E2E
SSE user events	PARTIAL	Endpoint exists
Voice/video calls	PARTIAL	WebRTC signaling code; BLOCKED browser test
Sylora AI call	PLACEHOLDER	/api/calls/sylora — no frontend caller

LIVE & streaming

Module	Class	Evidence
Create live room	REAL	POST /api/live → 201 verified
Live list / chat API	REAL	GET endpoints verified
WebRTC viewer/host	PARTIAL	Signaling implemented; no E2E media test
TURN / NAT	BLOCKED	SYLORA_ICE_SERVERS_JSON empty → STUN-only
RTMP / OBS ingest	MISSING	Only browser-source overlay + companion bridge
Recording	PARTIAL	Studio canvas record client-side only
Battles / resonance	PARTIAL	API + store hooks
Live Following tab	PLACEHOLDER	Empty state — no backend
Creator insights	BLOCKED	Needs OpenAI for AI portions

Media

Module	Class	Evidence
Video upload	PARTIAL	Endpoint exists; ffmpeg required
HLS transcode	PARTIAL	spawn ffmpeg in-process
Clips / video hub UI	STATIC UI	Lists from store

Gifts & wallet

Module	Class	Evidence
Gift catalog	REAL	GET /api/gifts — 10 tier gifts
Send gift + ledger	REAL	POST send 201 in runtime test (JSON mode)
LUMEN currency	MOCK	Labeled TEST; 10000 on register; not real money
Postgres atomic wallet	PARTIAL	Tested in pg-mem only
Gift GPU WebGL	PARTIAL	Procedural Three.js meshes; tests pass
Gift V2 Phoenix	PARTIAL	phoenix-preview.html works standalone
Gift atlas PNG	STATIC UI	2D sprite fallback in gift-engine
MediaPipe segmentation	PLACEHOLDER	Optional window global; not bundled

Sylora AI

Module	Class	Evidence
Text chat	BLOCKED	503 without OPENAI_API_KEY
Tool calling	PARTIAL	Code complete; untested without provider
Memory CRUD	REAL	API verified without OpenAI
Pending actions	PARTIAL	confirm/cancel routes exist
Voice realtime	BLOCKED	Needs OpenAI + mic permission
Browser STT/TTS	PARTIAL	Speech API used in UI when available
Living Sylora engine	PARTIAL	In-memory emotion/memory classes
Live co-host AI	BLOCKED	Needs OpenAI hook in ecosystem
Translation	BLOCKED	Provider keys missing
150+ ecosystem AI endpoints	MOCK/STATIC	Backend-only, no UI

Avatar (Living Sylora)

Module	Class	Evidence
3D rigged model	MISSING	No GLB/GLTF in repo
PNG sprite avatar	REAL	assets/sylora-avatar-v2-base.png + gesture PNGs
CSS blink / saccade	REAL	scheduleSyloraLife() in app.js
Viseme lip sync	PARTIAL	CSS sprite sheet from audio bands — not true lipsync
Emotions	STATIC UI	Expression PNG atlases exist unused in main AI view
Hands/body rig	NOT SUPPORTED	PNG rig assets exist but not animated skeletal

Business / education / science

Module	Class	Evidence
Business directory	PARTIAL	CRUD in JSON store
Org workspace (CRM, invoices)	PARTIAL	Ecosystem service — JSON backed
Courses / lessons	PARTIAL	api.test.mjs enrollment flow
Paid courses	BLOCKED	PAYMENT_PROVIDER_REQUIRED
Science calculators/tools	PARTIAL	Many POST endpoints; sparse UI
Conferences WebRTC	PARTIAL	Same as calls

Developer / agents

Module	Class	Evidence
API keys	PARTIAL	POST keys works in ecosystem
OAuth/OIDC	PLACEHOLDER	OAUTH_DOC JSON only
Agent marketplace	PARTIAL	UI + list endpoints

Monetization

Module	Class	Evidence
Wallet display	MOCK	TEST LUMEN
Stripe/payments	MISSING	env BLOCKED_EXTERNAL
Subscriptions	MISSING	No implementation
Creator payouts	MOCK	earnings field in wallet; no fiat

Infrastructure

Module	Class	Evidence
Docker compose	PARTIAL	compose.yaml valid syntax; Docker not on audit VM
CI/CD	MISSING	No .github/workflows
Monitoring	MISSING	Console logs only
Backups	MISSING	No strategy in repo

Summary counts

Class	~Count
REAL	12
PARTIAL	45
MOCK	4
STATIC UI	8
PLACEHOLDER	6
BROKEN	1 (favicon 404 — trivial)
MISSING	15
BLOCKED	14

Appendix: SYLORA_DUPLICATION_REPORT

Forensic scan 2026-08-13. Format: **A** → **duplicates B** → **keep** → **remove/merge**

Pages & navigation

A	B	Keep	Remove/merge
<code>renderProfile()</code>	<code>renderProfileLegacy()</code>	<code>renderProfile()</code>	Delete <code>renderProfileLegacy()</code> (~dead code, app.js)
Profile stats on <code>/profile</code>	Wallet on header + <code>/gifts</code> + ledger section	Single wallet entry in profile; header as shortcut only	Reduce repeated LUMEN displays
<code>/more</code> settings grid	Left rail "Налаштування"	Both OK (hub vs primary nav)	Clarify IA labels only
<code>/videos</code>	<code>/clips</code>	Both (different format)	Shared uploader component (currently duplicated <code>renderClipUploader /</code> <code>renderVideoUploader</code>)
Business org workspace	Business hub cards	Workspace for depth	Hub as index only
Learning courses	Science hub tools	Distinct products	Shared "create course" flow already duplicated in Create Hub

Wallet / monetization UI

A	B	Keep	Remove/merge
Header balance button	Profile vitals LUMEN	Header shortcut	—
<code>/gifts</code> balance hero	Profile wallet card	Profile as source of truth	Gifts page focus on catalog/send only
<code>/api/ledger</code> on profile	Gift send responses	Both APIs	—

Settings / profile

A	B	Keep	Remove/merge
/profile locale in form	Header #localeSwitch	Header (global)	Remove locale from profile form or sync one control
/security memory center	/ai memory tab	Security for privacy; AI for chat context	Document boundary; merge UI later
/identity	/profile bio/display	Identity = professional; profile = social	OK if documented; currently overlapping fields

Navigation components

A	B	Keep	Remove/merge
Left rail .nav	Mobile dock .nav	Both	Extract shared NAV_ITEMS config (currently duplicated in index.html)
Create Hub actions	Feed horizon buttons	Create Hub	—
Command palette routes	More grid	Palette for power users	—

CSS / design system

A	B	Keep	Remove/merge
design-consolidation.css	design-v2.css ... design-scenes-v6.css	One canonical token file + scene overrides	Merge 8 design-* files (all loaded in index.html)
styles.css	modules.css	Base + modules	Audit redundant rules
design-avatar-assembled.css	design-living-horizon.css	Consolidate avatar/horizon	
Hero background sylora-horizon-v3.png	Repeated in v4/v5/v6 CSS	Single --hero-bg variable	

Gift systems (major)

A	B	Keep	Remove/merge
gift-v2/* (canonical per gift-runtime.js)	gift-engine.js (2D canvas)	V2 + runtime router	Demote engine to fallback only
gift-gpu-engine.js (WebGL procedural)	gift-v2 WebGL renderer	GPU for v1 tier IDs; V2 for phoenix	Document matrix in one module
gift-sfx.js	gift-v2 physical-audio	V2 audio director	Merge SFX paths
PNG atlas sylora-gift-atlas-v1.png	WebGL meshes	Atlas for low-end fallback	OK dual path
Phoenix preview page	Live gift stage	Runtime controller	Preview dev-only

Backend / persistence

A	B	Keep	Remove/merge
JSON store.data.*	Postgres repositories	Postgres in production	JSON dev-only; stop dual-write paths in server.mjs
store.notify()	authSocial.createNotification()	Postgres path when enabled	Single notify abstraction
Live engagement in JSON	liveRepo.engagement()	Postgres when configured	Already branched — simplify
Ecosystem in JSON blobs	PostgresEcosystemRepository	Postgres for durable personal AI	Migration incomplete for all entities

API logic

A	B	Keep	Remove/merge
/api/search	/api/search/universal	Universal when authed	Deprecate duplicate result shaping
/api/ai/chat	/api/ai/command	Chat for AI screen; command for palette	Shared runSyloraAi() already — OK
/api/live/:id/resonance	/api/live/battles	One battle API	Two creation paths
Conference AI POST	Sylora AI chat	Shared OpenAI client	OK backend

State & services

A	B	Keep	Remove/merge
state.me	Cached users in store	Session from API	—
ecosystem/service.mjs (3885 LOC)	Domain files unused by UI	Split by bounded context	Extract only used domains

Assets

A	B	Keep	Remove/merge
sylora-expressions-v1.png	sylora-expressions-v2.png	v2 if used	Remove unused v1
sylora-visemes-v1.png	sylora-visemes-v2.png	v2 in CSS	Remove v1 if unreferenced
sylora-assistant-v1.png	sylora-avatar-v2-base.png	v2 for AI hero	v1 only in hero CSS ghosts
Multiple hand rig PNG sets v1–v4	Avatar motion in app.js	v4 if referenced	Audit ~45MB asset folder

Tests / docs

A	B	Keep	Remove/merge
docs/audit/CURRENT_STATE.md	This forensic audit	SYLORA_FULL_AUDIT.md	Archive old audit docs
docs/FINAL_IMPLEMENTATION_REPORT.md	Reality	Delete or mark historical	Misleading if read as current

Priority merge order (remediation reference only — not executed)

1. CSS design files → single design system
2. Gift runtime → one entry (`gift-runtime.js` already declared canonical)
3. JSON vs Postgres → production path only in prod
4. Dead `renderProfileLegacy`
5. Clip/video uploader duplication

Appendix: SYLORA_SECURITY_AUDIT

Date: 2026-08-13 | Mode: static + runtime (JSON dev server)

No secret values are printed below.

Executive summary

SYLORA implements meaningful baseline security (script passwords, hashed session tokens, CSP, rate limits, AI memory secret rejection). It is **not production-safe** as-is due to: dual persistence complexity, missing OAuth/payment hardening, admin/report data in JSON file, no CSRF tokens for cookie-less API (Bearer-only mitigates), and large attack surface from 296 endpoints.

Findings by severity

P1 — Critical

ID	Issue	Location	Evidence
S-P1-1	Production requires Postgres+Redis but dev JSON mode allows full data in plaintext file	<code>data/sylora.json</code> , <code>SYLORA_DATA_FILE</code>	File contains password hashes, sessions; committed path in repo <code>.gitignore</code> only
S-P1-2	No Google OAuth despite user-facing product expectation — phishing gap if added without PKCE	<code>integrations.mjs</code>	BLOCKED_EXTERNAL only
S-P1-3	Admin reports stored in same JSON store as user content — no RBAC separation in dev mode	<code>server.mjs</code> <code>/api/admin/*</code>	Works but file-level access = full compromise
S-P1-4	Media served from local disk without auth on GET <code>/media/:id</code>	<code>serveMedia()</code>	Any guessable UUID leaks uploaded video

P2 — Major

ID	Issue	Location
S-P2-1	Bearer token in localStorage (<code>sylora_token</code>) — XSS steals session	<code>public/app.js</code> bootstrap
S-P2-2	CORS not explicitly configured (same-origin SPA — OK today; risky if split)	<code>server.mjs</code>
S-P2-3	Rate limit falls back to in-memory per-process — bypassed under horizontal scale without Redis	<code>allowRequest()</code>
S-P2-4	AI tool actions expire 24h but no user notification channel audit	<code>aiCreateAction()</code>
S-P2-5	Browser-source overlay token in query string — leak via logs/referrer	<code>/obs-overlay.html?token=</code>
S-P2-6	Idempotency key required for Postgres gifts but not JSON gift path	POST <code>/api/gifts/send</code>
S-P2-7	Session locale PATCH accepts only uk/pl/en server-side but UI offers 14 languages	PATCH <code>/api/me</code> vs i18n

P3 — Minor

ID	Issue	Location
S-P3-1	favicon 404	public/
S-P3-2	CSP allows 'unsafe-inline' styles	securityHeaders()
S-P3-3	Logout succeeds without token (200)	runtime test
S-P3-4	.env.example contains dev postgres password placeholder	not a leak but weak default

Category checklist

Area	Status	Notes
Exposed secrets in repo	PASS	<code>.env.example</code> placeholders only; test keys in tests only
Auth bypass	PARTIAL	<code>requireUser</code> on protected routes; optional session on reads
IDOR	PARTIAL	Media GET unauthenticated; live SSE public
XSS	PARTIAL	<code>esc()</code> used in templates; <code>innerHTML</code> with escaped data
CSRF	N/A	Bearer header API — no cookies
SQL injection	PASS	Parameterized pg queries in repositories
Injection (JSON body)	PARTIAL	<code>safeText</code> truncation
WebSocket security	N/A	SSE instead
Upload safety	PARTIAL	Magic bytes + size limit; no AV scan
PII in logs	PARTIAL	Client errors posted to <code>/__client_error</code> truncated
Dependency vulns	PASS	npm audit 0 at audit time
AI prompt injection	PARTIAL	Safety filter in <code>living-sylora</code> ; tool allowlist
Memory secrets	PASS	<code>sanitizeMemoryValue</code> rejects api keys (test)

Exposed credentials scan

Pattern	Found in repo?
Live API keys	No
Private keys PEM	No (placeholder env only)
Hardcoded passwords	Test fixtures only
Session tokens in data file	Hashed (tokenHash) in test data from runtime registrations

Recommendations (document only — not implemented)

1. Authenticate media downloads or use signed URLs
2. Move session to httpOnly cookie + CSRF if same-site
3. Enforce Redis rate limits in production health gate (already in `/api/ready`)
4. Complete OAuth with PKCE before any UI
5. Strip query tokens from OBS overlay (POST token exchange)
6. Align locale UI with server allowed set

Appendix: SYLORA_PRODUCTION_READINESS

Assessment date: 2026-08-13 | Environment: Cloud audit VM

Readiness score: 18 / 100

Checklist

Capability	Status	Evidence
Environment separation	PARTIAL	NODE_ENV checks; <code>.env.local</code> load
Docker	PARTIAL	Dockerfile + compose.yaml; Docker not installed on audit VM
CI/CD	MISSING	No <code>.github/workflows</code>
Secrets management	PARTIAL	Env vars; no Vault/KMS
DB migrations	PARTIAL	<code>scripts/migrate.mjs</code> + SQL migrations; not run live
Backups	MISSING	No backup scripts
Logging	PARTIAL	<code>console.error</code> ; client error beacon
Metrics	MISSING	<code>/api/ecosystem/metrics</code> admin only
Tracing	MISSING	—
Health checks	REAL	<code>/api/health</code> , <code>/api/ready</code> verified
Error monitoring	MISSING	No Sentry/etc.
Rate limiting	PARTIAL	Implemented; Redis optional
CDN	MISSING	Static from Node
Object storage	MISSING	Local <code>data/media/</code>
Deployment script	PARTIAL	<code>scripts/deploy-prod.sh</code> for Hetzner
Rollback	MISSING	—
Disaster recovery	MISSING	—
Privacy (GDPR)	PARTIAL	<code>/api/privacy/requests</code> ecosystem route
Account deletion	MISSING	No route found
Data export	PARTIAL	<code>/api/ai/memory/export</code> only
HTTPS	PARTIAL	nginx example <code>infra/nginx/sylora.conf.example</code> ; HSTS opt-in
Horizontal scale	PARTIAL	Redis fanout designed; JSON store blocks scale
ffmpeg dependency	REAL	Present on VM; required for HLS
Production AI	BLOCKED	OPENAI_API_KEY required
Payments	BLOCKED	PAYMENT_PROVIDER
TURN servers	BLOCKED	SYLORA_ICE_SERVERS_JSON

`/api/ready` **behavior**

When `NODE_ENV=production` , `ready=false` unless postgres+redis+outbox all configured (`dependencyHealth()` in `server.mjs`).

Dev mode (audit): `ready=true` with JSON fallback.

Docker compose (static validation)

`docker compose config` — **not run** (docker missing). YAML structure reviewed: postgres healthcheck, redis AOF, migrate on start, healthcheck on sylora service.

Deployment path (documented)

1. `scripts/deploy-prod.sh` → git pull + docker compose up
2. `docs/DEPLOY-HETZNER.md` — manual VPS guide

Not executed in audit.

Blockers to production launch

1. Configure Postgres + Redis + run migrations
2. Set OPENAI_API_KEY (if AI is launch-critical)
3. Configure TURN for WebRTC
4. Replace TEST LUMEN with real payment provider or remove monetization claims
5. Add CI pipeline (test + lint + build)
6. Media storage strategy (S3-compatible)
7. Remove or secure dev JSON persistence in production images
8. Observability stack

What would pass a minimal prod bar

- Health/ready endpoints
- Security headers + CSP
- Password hashing
- Docker packaging exists
- Migration scripts exist
- Rate limit + body size limits

What fails a full prod bar

- No automated deploy verification in repo
- No backup/restore
- Monolithic ffmpeg transcode in API process
- 45MB static assets served by Node
- Single `app.js` / `service.mjs` god-files
- ~150 API endpoints without authz audit

Appendix: SYLORA_REMEDIATION_PLAN

Do not execute until approved. Ordered by dependency and readiness impact.

Phase 0 — P0 blockers

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P0	Dev JSON persistence in prod	server.mjs, compose.yaml	Fail boot if production without DATABASE_URL	Postgres	/api/ready 503 without DB in production
P0	Docker/CI absent	.github/workflows	Add CI: lint, test, compose config	—	PR checks green
P0	AI core blocked	.env, server.mjs	Document required OPENAI_API_KEY; degrade UI honestly	OpenAI account	Chat returns 200 with key
P0	WebRTC NAT	rtc-config.mjs	Configure TURN in SYLORA_ICE_SERVERS_JSON	TURN provider	Two-browser live test connects

Phase 1 — Architecture & foundations

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	Dual persistence confusion	store.mjs, repositories/*	Single write path per entity	Phase 0	No duplicate live/chat in JSON when PG on
P1	3885 LOC service.mjs	ecosystem/service.mjs	Extract domains with tests	—	≤500 LOC per module
P1	11 CSS files	public/design-*.css	Merge to design-tokens + scenes	Design review	2 CSS files loaded
P1	Unauthenticated media	server.mjs serveMedia	Signed URLs or auth check	—	401 without token

Phase 2 — Core user flows

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	No browser E2E tests	tests/	Playwright: register → post → logout	CI	1 green E2E pipeline
P1	Google auth missing	server.mjs, app.js	OAuth PKCE or remove from marketing	Google console	Login with Google works
P2	Password recovery	server.mjs	Email reset flow	SMTP	Reset link works
P2	Live Following empty	live repo	Implement follow graph for live	Social graph	Tab shows rooms
P2	Clip/video uploader dup	app.js	Shared upload component	—	One uploader module

Phase 3 — Sylora AI / avatar

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	AI chat blocked	server.mjs, OpenAI	Configure provider + eval tests	OPENAI_API_KEY	ai.test passes live
P2	Avatar is PNG not 3D	app.js, assets	Decide: honest 2D OR import GLB	Art pipeline	Product truth in UI copy
P2	Viseme fake lipsync	app.js setSyloraViseme	Integrate viseme v2 sheet or drop claim	—	Sync within 100ms of audio
P2	150 orphan AI endpoints	ecosystem/routes.mjs	Hide or wire UI	Product spec	Each endpoint has UI or @internal

Phase 4 — Live / social / creator

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	WebRTC E2E unverified	app.js live/calls	Manual + automated WebRTC test	TURN	Host+viewer video
P2	RTMP absent	—	Scope decision: browser-only or ingest	Infra	Document limitation
P2	Companion not bundled	companion.mjs	Document start script in compose	OBS	Studio connects
P3	Battle dual API	server.mjs, routes.mjs	Unify battle creation	—	One endpoint

Phase 5 — Monetization

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	TEST LUMEN misleading	app.js, wallet	Rename to DEMO or integrate Stripe	Payment provider	Real checkout OR honest label
P1	Paid courses blocked	server.mjs enroll	Payment webhook	Stripe	Paid enroll works
P2	Creator payouts	wallet repo	Payout ledger + admin	Legal	Export CSV

Phase 6 — Responsive / design consistency

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P2	Tablet not tested	modules.css	Test 768–1024 breakpoints	—	Screenshot suite
P2	Mixed EN/UK in nav	i18n.js, index.html	Full i18n for nav labels	—	PL/EN complete
P3	favicon 404	public/	Add favicon.ico	—	200 on /favicon.ico

Phase 7 — Security / performance / testing

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P1	localStorage token	app.js	httpOnly cookie migration	—	XSS token theft mitigated
P2	45MB assets	public/assets	Compress, lazy-load, CDN	CDN	LCP < 2.5s
P2	Gift runtime paths	gift-runtime.js	Consolidate fallback chain	Phase 1 CSS	One play() API
P3	Dependency audit in CI	package.json	npm audit --production	CI	Fail on high

Phase 8 — Production readiness

PRIORITY	PROBLEM	FILES/MODULES	ACTION	DEPENDENCIES	ACCEPTANCE CRITERIA
P0	No backups	infra/	pg_dump cron + media sync	Postgres	Restore drill
P1	No metrics	server.mjs	Prometheus /health metrics	—	Dashboard
P1	ffmpeg in API process	server.mjs	Queue workers for transcode	Redis queue	API stays responsive
P2	Account deletion	routes.mjs	GDPR delete user	Legal	Data removed in 30d

Top 10 tasks by readiness ROI

1. Postgres+Redis production boot + migrations
2. OPENAI_API_KEY + Sylora chat E2E
3. TURN for Live/Calls
4. CI (test/lint/build)
5. Playwright smoke E2E
6. Payment provider OR remove TEST LUMEN branding
7. CSS/design consolidation
8. Media auth/signed URLs
9. Browser WebRTC verification suite
10. favicon + honest AI/avatar product copy